

Experiment No.: 03

Title: Basic Image Transformation



Batch: A2 Roll No.: 16014324028

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Aim: Learn and perform rotation, translation, scaling, and flipping algorithms

Resources needed: Python 3.x, OpenCV (pip install OpenCV-python)

Pre Requisite: Students should have prior knowledge of basic concept of Computer Vision, Data sets required and image transformation algorithms

Theory: Methods of transformation

Procedure:

1. Load a color image.
2. Convert to Gray Scale
3. Apply Transformation techniques
4. Display it.

Results: (Steps with screenshots)

1. Prepare a document with all Steps with screenshots

Questions:

1. What is the difference between a linear transformation and an affine transformation in image processing?

Ans:- A linear transformation is a transformation that maps a point from one coordinate system to another while preserving the origin. It can be represented using a matrix and includes operations such as rotation, scaling, and shearing. An affine transformation is a more general form of transformation that combines a linear transformation with translation. Therefore, affine transformations can perform rotation, scaling, shearing, and translation. Mathematically, a linear transformation can be written as $P' = AP$, whereas an affine transformation is written as $P' = AP + t$, where A is the transformation matrix and t represents translation.

2. Why do we often use homogeneous coordinates for rotation, scaling, and translation in computer vision?

Ans:- Homogeneous coordinates are commonly used because they allow rotation, scaling, and translation to be represented using a single matrix representation. In two-dimensional Cartesian coordinates, rotation and scaling can be represented using a 2×2 matrix, but translation cannot be directly represented in the same form. By representing a point as $(x, y, 1)$, a 3×3 homogeneous

transformation matrix can include rotation, scaling, and translation together. This makes it easier to combine multiple transformations and apply them efficiently in computer vision and image processing.

3. How does the choice of rotation centre affect the resulting transformed image?

Ans:- The rotation centre determines the point around which all the pixels of an image are rotated. When an image is rotated around its centre, the image generally remains positioned around the same location, although its orientation changes. If the rotation is performed around another point, such as a corner, the image moves around that point and parts of the image may shift outside the displayed region. Therefore, changing the rotation centre changes both the orientation and the position of the transformed image.

4. Explain why scaling by a factor less than 1 may cause loss of image information.

Ans:- Scaling an image by a factor less than 1 reduces its width and height, which also reduces the total number of pixels available to represent the image. For example, scaling an image by 0.5 reduces both its width and height to half, resulting in only one-fourth of the original number of pixels. During this process, multiple original pixels may be combined or some pixel-level details may be discarded through interpolation. As a result, fine details and high-frequency information can be lost, and increasing the image size again cannot completely recover the original information.

5. What happens to an image's pixel coordinates when applying a rotation of 90° vs 180° ?

Ans:- When a point (x,y) is rotated by 90° counterclockwise around the origin, its coordinates become $(-y,x)$. This means that the x and y coordinates effectively exchange their roles, with one coordinate changing sign. For a 180° rotation, the coordinates become $(-x,-y)$, meaning both coordinate directions are reversed. In an actual image, the transformation is performed relative to the selected rotation centre and the image dimensions, but the fundamental coordinate transformations remain the same. Thus, a 90° rotation changes the orientation by a quarter turn, while a 180° rotation turns the image completely upside down.

Numerical Example

1: Show mathematically why rotation matrices preserve distances between points.

2: *Scaling in image transformation*

Outcomes:

Code:-

```
import cv2
import matplotlib.pyplot as plt

img =
cv2.imread("/media/galdrux/galdrux_storage/sem_5/CV/dataset/satellite_9.jpg")
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)

h, w = img.shape[:2]

rotation_matrix = cv2.getRotationMatrix2D((w // 2, h // 2), 45, 1)
rotated = cv2.warpAffine(img, rotation_matrix, (w, h))

translation_matrix = cv2.getRotationMatrix2D((0, 0), 0, 1)
translation_matrix[0, 2] = 100
translation_matrix[1, 2] = 50
translated = cv2.warpAffine(img, translation_matrix, (w, h))

scaled = cv2.resize(img, None, fx=0.5, fy=0.5)

flipped_horizontal = cv2.flip(img, 1)
flipped_vertical = cv2.flip(img, 0)

plt.figure(figsize=(12, 8))

plt.subplot(2, 4, 1)
plt.imshow(img)
plt.title("Original")
plt.axis("off")

plt.subplot(2, 4, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

plt.subplot(2, 4, 3)
```

```
plt.imshow(rotated)
plt.title("Rotation 45°")
plt.axis("off")

plt.subplot(2, 4, 4)
plt.imshow(translated)
plt.title("Translation")
plt.axis("off")

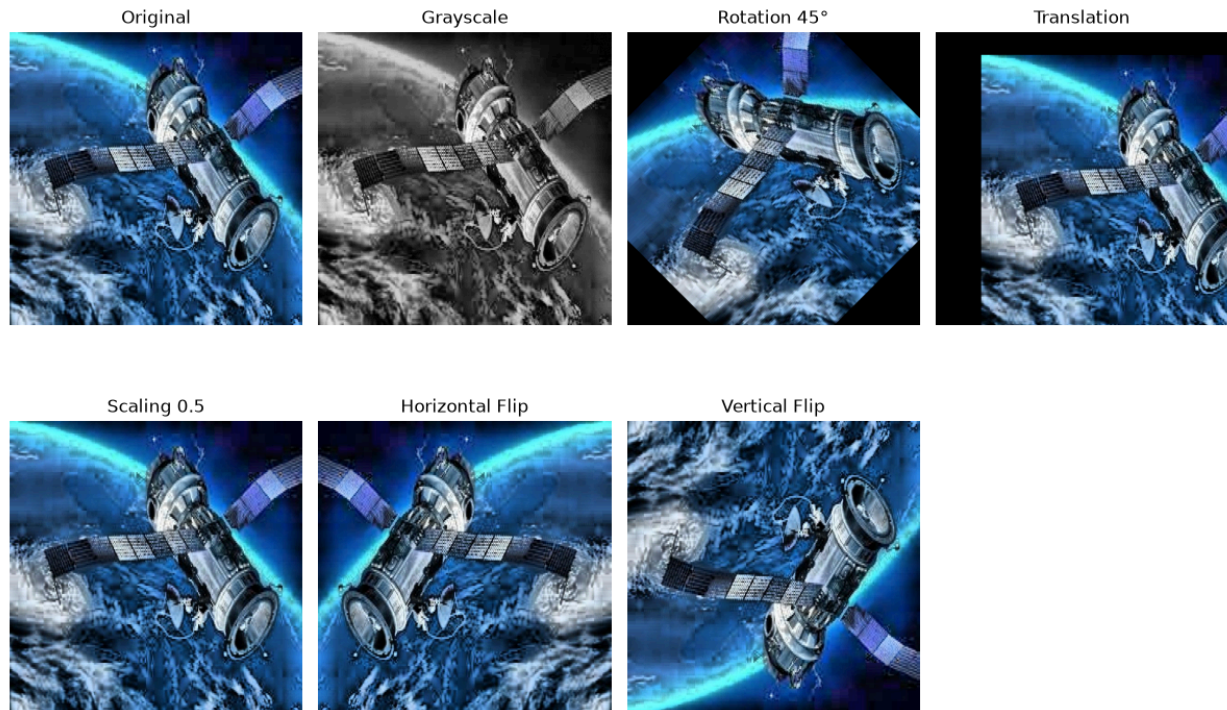
plt.subplot(2, 4, 5)
plt.imshow(scaled)
plt.title("Scaling 0.5")
plt.axis("off")

plt.subplot(2, 4, 6)
plt.imshow(flipped_horizontal)
plt.title("Horizontal Flip")
plt.axis("off")

plt.subplot(2, 4, 7)
plt.imshow(flipped_vertical)
plt.title("Vertical Flip")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:-



Conclusion: (Conclusion to be based on the objectives and outcomes achieved)

The experiment successfully demonstrated the basic image transformation techniques of rotation, translation, scaling, and flipping using Python and OpenCV. The effects of different transformations on image coordinates and dimensions were observed and analyzed. The experiment also provided an understanding of linear and affine transformations, homogeneous coordinates, rotation centres, and information loss during scaling.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References

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Websites: